



Project Purposal

CTF Platform



Team Members

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Introduction

The 'Death Note CTF' platform is an interactive Capture the Flag environment designed for students and cybersecurity enthusiasts to practice real-world security challenges. It covers multiple domains such as cryptography, web attacks, OSINT, and forensics, special offering hands-on experience in identifying vulnerabilities and extracting hidden information.

Problem Statement

Current cybersecurity education often relies on theoretical concepts, lacking practical and engaging platforms for students to learn through hands-on problem solving. Existing CTF platforms are either too advanced for beginners or not comprehensive enough to cover different areas of cybersecurity. This project addresses this gap by providing a thematic, gamified, and beginner-friendly yet challenging platform that simulates real-world attack and defense scenarios.

Challenge Category

1. *Crypto:*

- Caesar Cipher,
- XOR,
- RSA Mini-Challenge,
- Vigenère Cipher,
- Playfair Cipher,
- Hashing tasks.
- Wiener's RSA

2. *OSINT:*

- Metadata extraction,
- social media clue hunting,
- hidden information discovery.
- hidden image discovery.

3. *Web Attacks:*

- SQL Injection, Union-based SQLi, blind SQLi,
- Reflected XSS, Stored XSS,
- Hidden directories,



- LFI,
- Union-based SQLi
- IDOR vuln
- weak credential brute force,
- directory enumeration.

4. Forensics:

- LSB Steganography (hidden messages in images).
- Image analysis (recover deleted files).
- PCAP analysis (Wireshark logs).
- Extract strings
- PDF Redaction
- DNS analysis

5. Specials:

- Wifi-handshake
- Tone dialing
- Spectrograms
- PDF Redaction
- DNS analysis

Progress Tracker:

- Gamified scoring system with live rankings.

Hints & Writeups:

- Optional hints to assist players and post-challenge writeups for learning.

Tools & Technology

Programming Language:

- python → Backend challenge logic
- Python (Flask) → Automation, scripting, web challenges.
- JS, HTML, CSS → Front-end

Database:

- Mini SQLi database for authentication, storage, web challenges.

Framework & Tools:

- Crypto: Custom-coded ciphers, OpenSSL, Hashcat.



- OSINT: Exiftool, Google Dorks, Sherlock.
- Web Attacks: Burp Suite, SQLMap, Hydra.
- Forensics: Steghide, Binwalk, Wireshark, Strings.

Platform:

- Linux (Kali) with Flask-based web UI (localhost) for challenge access.